# Pre-analysis Plan

Many Daughters Project

## H1. Having daughters leads to stronger support for gender equality (as measured by attitudes)

### 1.1 Variables selected

Table 1

| | Variable | Data availability | Current scale | Variable transformation |
|---|---|---|---|---|
| Dependent variables measuring attitudes on gender equality (X) | (plj0620) democracy: women have the same rights as men | 2016-2018 | 0-10 scale (0: should not happen in a democracy, 10: should definitely happen in a democracy) | N/A |
| | (plj0621) position of women: working as women (Having a job is the best way for a woman to be independent. | 2016-2018 | 1-7 likert scale (1: strongly disagree & 7: strongly agree) | N/A |
| | (plj0622) position of women: working as married women (Even a married woman should have a paid job so that she can be financially independent.) | 2016-2018 | 1-7 likert scale (1: strongly disagree & 7: strongly agree) | N/A |
| | (plj0623) position of women: when woman earns more than partner, problems | 2016-2018 | 1-7 likert scale (1: strongly disagree & 7: strongly agree) | Reverse code |
| | (plj0624) position of women: education of son more important than of daughter | 2016-2018 | 1-7 likert scale (1: strongly disagree & 7: strongly agree) | Reverse code |
| | (plj0625) position of women: at home, husband should have the final say. | 2016-2018 | 1-7 likert scale (1: strongly disagree & 7: strongly agree) | Reverse code |
| Main independent variable | (pld0025) daughters present | 2001, 2006, 2011, 2016 | 1: Yes, 0: No | N/A |
| | (pld0026) number of daughters | 2001, 2006, 2011, 2016 | From 1 onwards | N/A |
| Control Variables | Age -> (gebjahr) Birth Year | 1984-2022 | 4 digit. In ppathl data | Age squared |
| | (sex) Gender | 1984-2022 | 1: male 2:female. In ppathl data | Recode to binary: male=1, & 0, female |
| | (pgfamstd) What is your marital status? | 1984-2022 | In pgen data. | Recode to binary: married (1/0), single (1/0) |

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|---|---|---|---|---|
| | (pgemplst) Employment status | 1984-2022 | In pgen data. | Recode to binary: full time employment=1, & 0, otherwise |
| | (pgbilzeit) Amount of education or training in years | 1984-2022 | In pgen data. | N/A |
| | (pglabgro) Individual Income: What did you earn from your work last month? Please state both: gross income, which means income befor deduction of taxes and social security, and net income, which means income after deduction o taxes, social security, and unemployment and health insurance. | 1984-2022 | Continuous. In pgen data. | N/A |
| | (hghinc) Household income: What was your total income from interest, dividends, profits, and profit distribution from all of your investments in the last calendar year? | 1984-2022 | Continuous. In hgen | N/A |
| | (plh0258) religion | 2015-2021 | Categorical. In pl | Construct religion dummies |
| | (fprofedu) – Level Of Education Father (or plg0244 in pl) | N/A | In bioparent | categorical |
| | (msedu) – Level Of Education Mother (or plg0245 in pl) | N/A | In bioparent | categorical |
| | (vnat) Nationality Father | N/A | In bioparent | Recode to binary: German(1/0) |
| | (mnat) Nationality Mother | N/A | In bioparent | Recode to binary: German(1/0) |
| | (pgnation) Nationality | N/A | In pgen | |

cid: original household number, pid: never changing person id , hid: current wave HH number, syear: survey year

## 1.2 Inclusion/Exclusion Criteria

Inclusion criteria:

- At least one child present ((pld0025) daughters present 1: Yes, 0: No)
- Survey year = 2016-2018 (due to data availability)

## 1.3 Exact analysis

We will estimate the effect of having daughters on parents' gender-equality attitudes with an ordinary-least-squares (OLS) regressions that cluster robust standard errors at the individual level (pid). We estimate,

$$Y_{ik} = \beta_0 + \beta_1 DaughterPresence_i + \beta_2 NumDaughters_i + \mathcal{Z}_i'\gamma + \epsilon_{ik}$$

Gender equality ($Y_{ik}$) is measured by six SOEP attitude items administered in 2016. We will estimate the model using a composite measure of gender equality using the mean of the six reverse-coded, standardized items. The six items are,

- (plj0620) democracy: women have the same rights as men
- (plj0621) position of women: working as women (Having a job is the best way for a woman to be independent.
- (plj0622) position of women: working as married women (Even a married woman should have a paid job so that she can be financially independent.)
- (plj0623) position of women: when woman earns more than partner, problem
- (plj0624) position of women: education of son more important than of daughter
- (plj0625) position of women: at home, husband should have the final say.

Key independent variables include,

- a binary indicator (pld0025) capturing whether the respondent has any daughters ($DaughterPresence_i$), and a
- continuous count variable (pld0026) records the exact number of daughters in the household ($NumDaughters_i$).

The coefficient $\beta_1$ therefore represents the average difference in gender-equality support between parents with versus without daughters, while $\beta_2$ traces the marginal impact of each additional daughter.

To isolate the daughter effect from confounding family and background characteristics, we include the following covariates for $\mathcal{Z}_i'$.

- Respondent's age (constructed from birth year) and its square
- Gender
- marital status (pgfamstd),
- full-time employment (pgemplst)
- years of education (pgbilzeit)
- individual and household income (pglabgro and hghinc)
- religion(plh0258)
- parental education and nationality (fprofedu, msedu, vnat, mnat)
- nationality (pgnation)

### 1.4 Interpretation of results

We will use the p-value of the coefficient to assess statistical significance. If the p-value is below 0.05, we reject the null hypothesis at the 5% level.

We will consider a positive and statistically significant coefficient on $NumDaughters_i$ (i.e., $\beta_2$) as evidence that having a greater number of daughters is associated with stronger support for gender equality.

## H2. Having daughters leads to more left-leaning political preferences.

### 1.1 Variables selected

Table 2

| | Variable | Data availability | Current scale | Variable transformation |
|---|---|---|---|---|
| Main dependen variable on left leaning politica views | (plh0004) Political attitude: In politics people o talk about “left” and “right” when it comes to characterize different political attitudes. If you t about your own political views: Where would y place yours? | 1984-2021 | 0-10 (0: far left & 10: far right) | Reverse code |

## 1.2 Inclusion/Exclusion Criteria

Inclusion criteria:

- At least one child present ((pld0025) daughters present 1: Yes, 0: No)
- Survey year = 2015-2021. Politics evolve over time alongside societal changes. Therefore, to analyze a question anchored in the context of 2025, it is most appropriate to focus on the most recent relevant political period. Since 2021, the last year of data collection, marks Angela Merkel’s final year as Chancellor, we base our analysis on observations from her last political mandate.

## 1.3 Exact analysis

We will estimate the effect of having daughters on parents’ political orientation with an ordinary-least-squares (OLS) regressions that cluster robust standard errors at the individual level (pid). We estimate,

$$Y_{\text{ik}} = \beta_0 + \beta_1 DaughterPresence_i + \beta_2 NumDaughters_i + \mathcal{Z}_i'\gamma + \epsilon_{ik}$$

Political orientation ($Y_{ik}$) is measured by one SOEP attitude items administered in 2016. This item is,

- (plh0004) Political attitude: In politics people often talk about “left” and “right” when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?

Key independent variables include,

- a binary indicator (pld0025) capturing whether the respondent has any daughters ($DaughterPresence_i$), and a
- continuous count variable (pld0026) records the exact number of daughters in the household ($NumDaughters_i$).

The coefficient $\beta_1$ therefore represents the average difference in left leaning political views between parents with and without daughters, while $\beta_2$ traces the marginal impact of each additional daughter.

To isolate the daughter effect from confounding family and background characteristics, we include the same covariates specified in Table 1, except of course household income, i.e.,

- Respondent’s age (constructed from birth year) and its square
- Gender
- marital status (pgfamstd),
- full-time employment (pgemplst)

- years of education (pgbilzeit)
- individual income (pglabgro)
- religion(plh0258)
- parental education and nationality (fprofedu, msedu, vnat, mnat)
- nationality (pgnation)

### 1.4 Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

We will use the **p-value** of the coefficient to assess statistical significance. If the p-value is below 0.05, we reject the null hypothesis at the 5% level.

We will consider a positive and statistically significant coefficient on $NumDaughters_i$ (i.e., $\beta_2$) as evidence that having a greater number of daughters is associated with stronger support for left leaning political views.

## H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

We conceptualise gender equality in the household through behavioural indicators and outcomes, such as equal participation in domestic tasks (e.g., meal preparation, laundry), shared caregiving responsibilities (e.g., caring for children), and equitable financial contributions (e.g., individual income, mortgage payments, utility bills).

However, after reviewing the available datasets—pl, hl, biol, biobirth, bioparen, pgen, hgen, and ppathl—we were unable to identify variables that capture these dimensions of household gender equality in a way that aligns with the time periods for which data on daughter presence and number of daughters are available**.**

We will instead use household income as a proxy for gender equality within the household. Generally speaking, a well-functioning household tends to involve two employed individuals, and higher income levels are often indicative of greater overall household functioning. Indeed, higher income should reflect better understanding of human relationships (relational skills, sharing of responsibilities), which in turn suggests a higher quality of couple dynamics. We do not distinguish whether the individual resides with the parent of their children or with another partner, treating both cases as similar instances of cohabitation. We also implicitly assume that living alone reflects a degree of dysfunction relative to the normative expectation of partnered living.

Inclusion criteria:

- At least one child present ((pld0025) daughters present 1: Yes, 0: No)
- Survey year = 2015-2021. Society evolves over time, we therefore examine a 2025 question in light of the most recent political period available in the questionnaire.

### 1.3 Exact analysis

We will estimate the effect of having daughters on gender equality in the household with an ordinary-least-squares (OLS) regressions that cluster robust standard errors at the individual level (pid). We estimate,

$$Y_{\text{ik}} = \beta_0 + \beta_1 DaughterPresence_i + \beta_2 NumDaughters_i + \mathcal{Z}_i'\gamma + \epsilon_{ik}$$

Gender equality in the household ($Y_{ik}$) is measured by one SOEP item administered in every year of the survey. This item is:

- hlc0005_h (net household income): If you look at the total income of all of the members of your household: what is your monthly household income today ? // Euros per month.

Key independent variables include,

- a binary indicator (pld0025) capturing whether the respondent has any daughters ($DaughterPresence_i$), and a
- continuous count variable (pld0026) records the exact number of daughters in the household ($NumDaughters_i$).

The coefficient $\beta_1$ therefore represents the average difference in household income between parents with and without daughters, while $\beta_2$ traces the marginal impact of each additional daughter.

To isolate the daughter effect from confounding family and background characteristics, we include the same covariates specified in Table 1, except household income, i.e.:

- Respondent's age (constructed from birth year) and its square
- Gender
- marital status (pgfamstd),
- full-time employment (pgemplst)
- years of education (pgbilzeit)
- individual income (pglabgro)
- religion(plh0258)
- parental education and nationality (fprofedu, msedu, vnat, mnat)
- nationality (pgnation)

### 1.4 Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

We will use the **p-value** of the coefficient to assess statistical significance. If the p-value is below 0.05, we reject the null hypothesis at the 5% level.

We will consider a positive and statistically significant coefficient on $NumDaughters_i$ (i.e., $\beta_2$) as evidence that having a greater number of daughters is associated with stronger support for left leaning political views.

## H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

### 1.1 Variables selected

Same as those listed in Table 1.

### 1.2 Inclusion/Exclusion Criteria

Inclusion criteria:

- At least one child present ((pld0025) daughters present 1: Yes, 0: No)
- Survey year = 2016 (due to data availability)

## 1.3 Exact analysis

We will estimate the effect of having daughters on gender equality is stronger for men than for women with an ordinary-least-squares (OLS) regression that cluster robust standard errors at the individual level (pid). We estimate,

$$Y_{\text{ik}} = \beta_0 + \beta_1 DaughterPresence_i + \beta_2 Male_i + \beta_3\Big(DaughterPresence_i \times Male_i\Big) + \beta_4 NumDaughters_i + \mathcal{Z}_i'\gamma + \epsilon_{ik}$$

Gender equality ($Y_{ik}$) is measured by six SOEP attitude items administered in 2016. We will estimate the model using a composite measure of gender equality using the mean of the six (reverse-coded) items. The six items are,

- (plj0620) democracy: women have the same rights as men
- (plj0621) position of women: working as women (Having a job is the best way for a woman to be independent.
- (plj0622) position of women: working as married women (Even a married woman should have a paid job so that she can be financially independent.)
- (plj0623) position of women: when woman earns more than partner, problem
- (plj0624) position of women: education of son more important than of daughter
- (plj0625) position of women: at home, husband should have the final say.

Key independent variables include,

- $DaughterPresence_i$ - a binary indicator (pld0025) capturing whether the respondent has any daughters
- Male (sex) – 1 for men, 0 for women.
- $DaughterPresence_i \times Male_i$ - interaction that captures how much larger the daughter effect is for men than for women.
- $NumDaughters_i$ records the exact number of daughters in the household

We include the following covariates for $\mathcal{Z}_i'$.

- Respondent's age (constructed from birth year) and its square
- marital status (pgfamstd),
- full-time employment (pgemplst)
- years of education (pgbilzeit)
- individual and household income (pglabgro and hghinc)
- religion(plh0258)
- parental education and nationality (fprofedu, msedu, vnat, mnat)
- nationality (pgnation)

## 1.4 Interpretation of results

We will use the **p-value** of the coefficient to assess statistical significance. If the p-value is below 0.05, we reject the null hypothesis at the 5% level.

We will consider a positive and statistically significant coefficient on $DaughterPresence_i \times Male_i$ (i.e., $\beta_3$) as evidence that having daughters leads to a larger increase in the support of gender equality for men than form women.